

FILIP

Fine-grained Interactive Language-Image Pre-Training

Learning word-patch alignment without bounding boxes

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Table of contents

01

Intro

What preceded FILIP and why it was needed.

02

How FILIP flipped the equation

The innovation it brought.

03

Dataset

04

Training

How it was trained and what was seen.

Table of contents

05 Results

How preceded FILIP and why it was needed.

06 Ablation Studies

More findings.

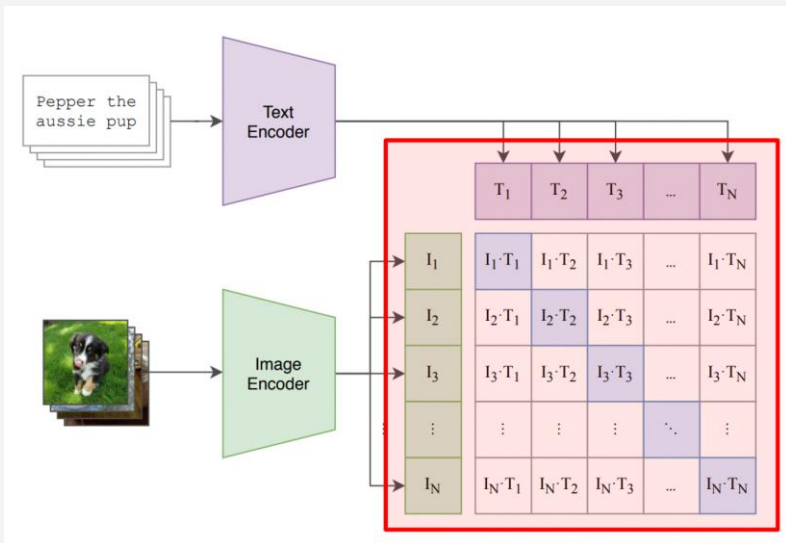
07 Visualization

Seeing the true form

01

Intro

How it all started



A quick question....

“How would a model know that ‘cat’ corresponds to the animal and ‘basin’ corresponds to the bowl — without any bounding box annotations?”

”

—Someone Famous



🐱 “cat” sitting in a “bowl”

What Problem are we solving?

“Vision-Language Alignment Problem”

VLMs usually only know the **image caption as a whole**, not where words correspond inside the image.

Goal:

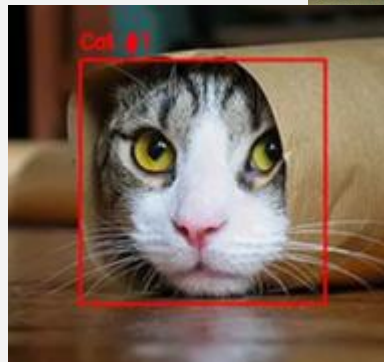
Learn relationship between images and text.

Example:

Image → "a cat in a basket"

Applications:

- Image search
- Image captioning
- Zero-shot classification



02

**How FILIP
flipped the
equation**

FILIP's Working

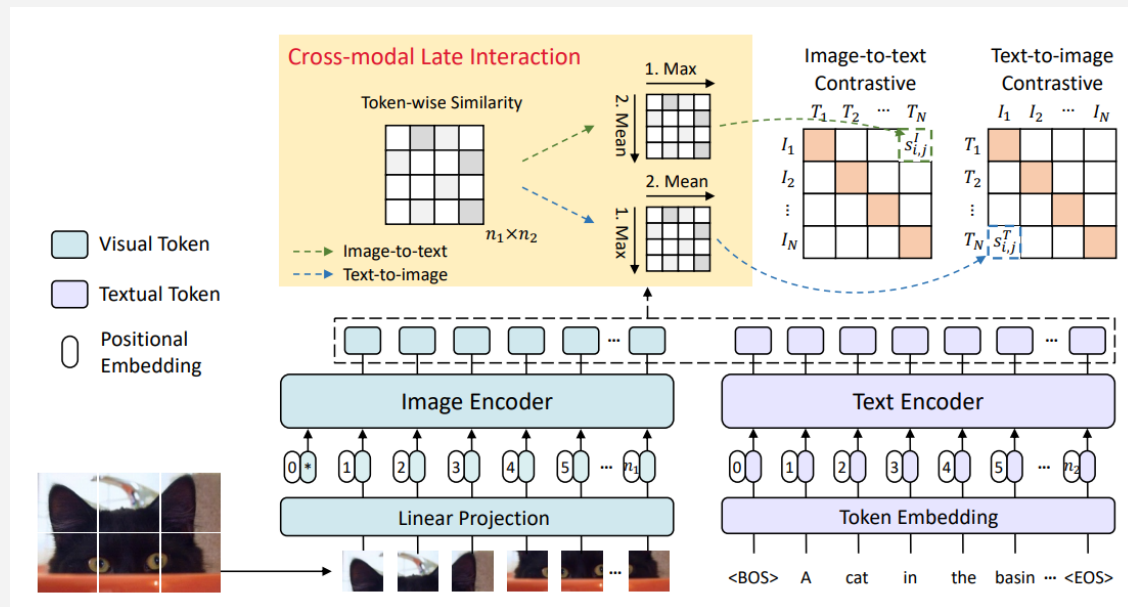
Instead of:
image $\leftrightarrow$ sentence

FILIP does:
image patches $\leftrightarrow$ words

Patch1 $\leftrightarrow$ word1

Patch2 $\leftrightarrow$ word2

Patch3 $\leftrightarrow$ word3



“FILIP aligns image patches with words instead of compressing everything into one vector.”

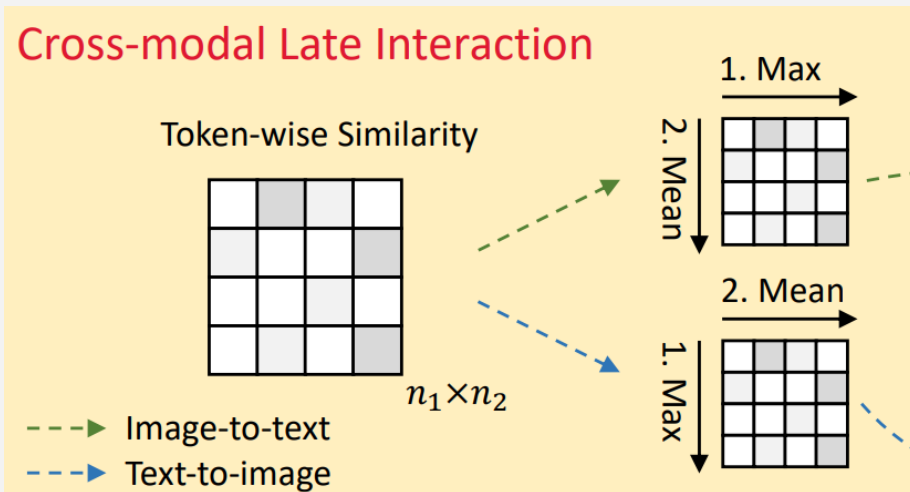
Method: Cross-Modal Late Interaction

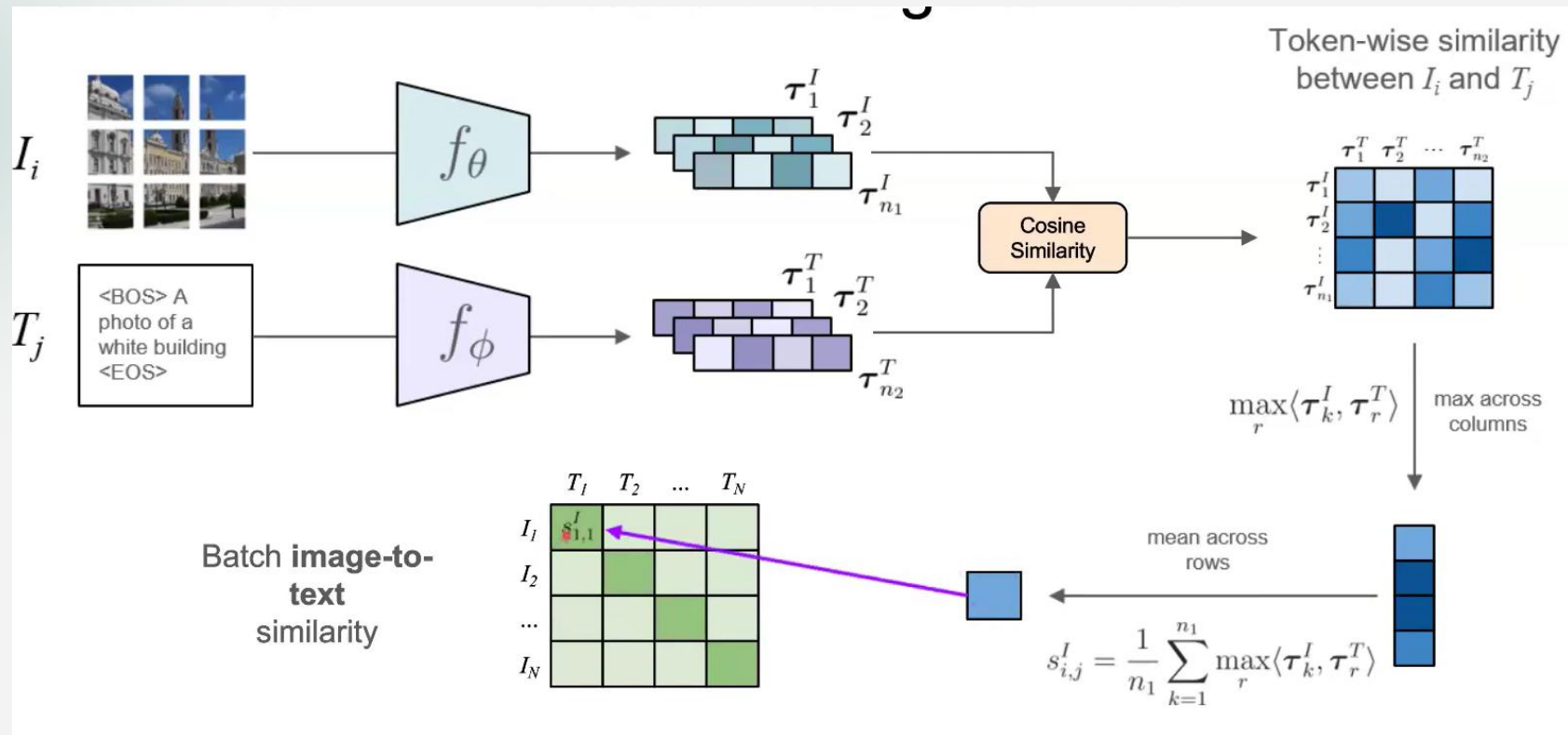
For each patch:
find most similar word
Then average.

Example matrix:

	cat	basin	in
patch1	✓		
patch2		✓	
patch3			

Patch chooses its best word.
Which is the **core contribution**.





Maths behind the Model

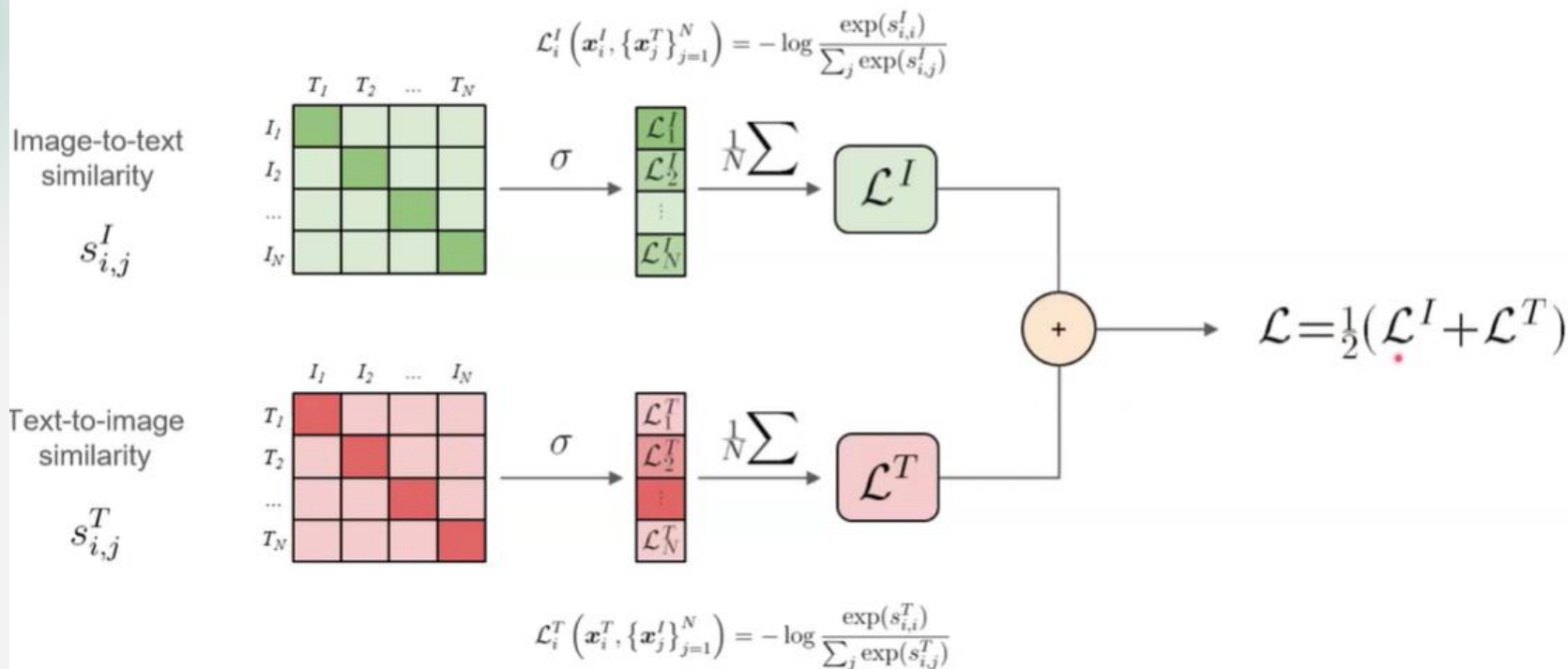
In each training batch, b image-text pairs $\{x_{Ik}, x_{Tk}\}$ where k is from 1 to b are sampled. For image x_{Ik} in image-text pair $\{x_{Ik}, x_{Tk}\}$, x_{Tk} is its positive, while the other texts will be used as in-batch negatives. The image-to-text contrastive loss $\mathcal{L}_{Ik}$ for x_{Ik} can then be formulated as:

$$\mathcal{L}_k^I(\mathbf{x}_k^I, \{\mathbf{x}_j^T\}_{j=1}^b) = -\frac{1}{b} \log \frac{\exp(s_{k,k}^I)}{\sum_j \exp(s_{k,j}^I)}$$

- where $s_{Ik,j}$ denotes the **similarity of the k -th image to the j -th text**.

Similarly, the **text-to-image contrastive loss** for x_{Tk} is:

$$\mathcal{L}_k^T(\mathbf{x}_k^T, \{\mathbf{x}_j^I\}_{j=1}^b) = -\frac{1}{b} \log \frac{\exp(s_{k,k}^T)}{\sum_j \exp(s_{j,k}^T)}$$



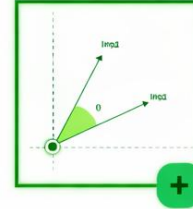
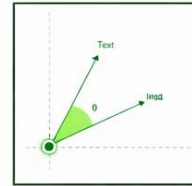
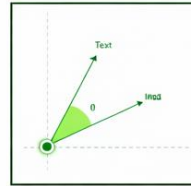
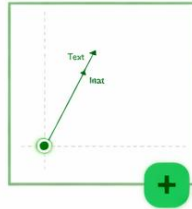
Maths behind the Model

The **total loss** of this mini-batch can be represented by:

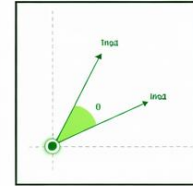
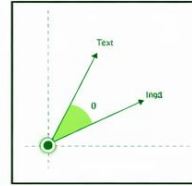
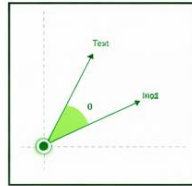
$$\mathcal{L} = \frac{1}{2} \sum_{k=1}^b (\mathcal{L}_k^I + \mathcal{L}_k^T)$$



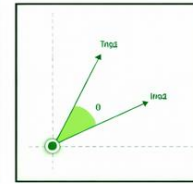
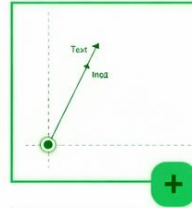
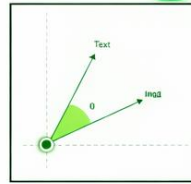
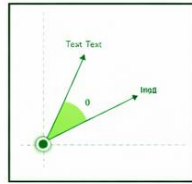
A cat curls up softly,
purring in the sunlight.



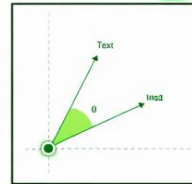
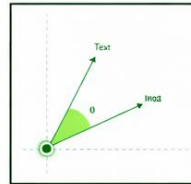
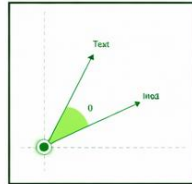
A dog wags its tail
with endless joy.



A bird soars freely
across the open sky.

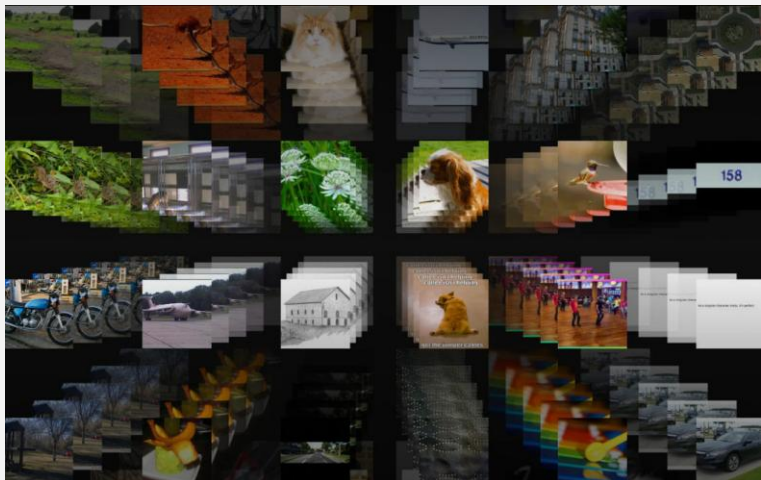


A fish glides gracefully
through the water.



cosine similarity = 1

Dataset



Datasets

FILIP300M dataset
~340M image-text pairs

30%

■ YFCC100M

5%

CC3M & CC12M ■

65%

■ Internet Scrapping

Filtering rules: remove small images, remove duplicates, English text only



04

Training

Methods Used

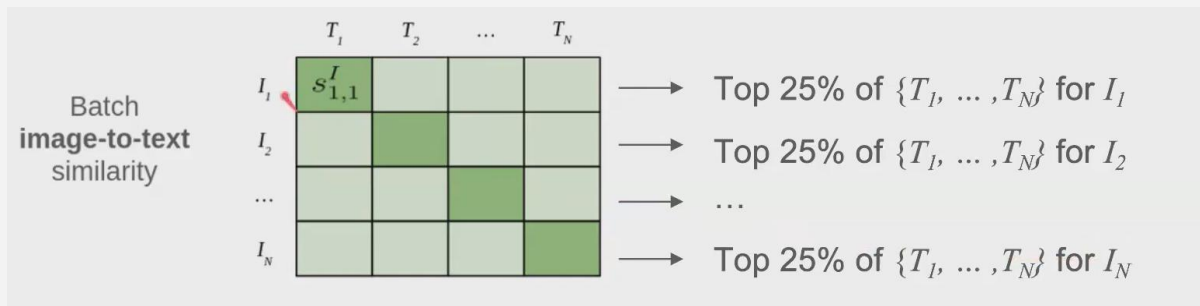
For C prompt templates, each label is augmented to C different texts $xT_1, xT_2, \dots, xT_C$.

AutoAugment for image augmentation.

Back Translation is used for text augmentation.

FILIP can be inefficient in terms of communication, memory and computation.

Only highest 25% tokens with max similarity taken.





I

"building"

Prompt
Ensemble

"A photo of a **building**.
It's common *in* daily life."

$\vdots$

"A JPEG of a **building**, I
like it."

T_1

T_C

Similarity

$$\frac{1}{C} \sum_{c=1}^C s^I(I, T_c)$$

$C = \#$ prompt templates



05

Results

Zero-Shot Classification

FILIP improves performance despite using **less data than CLIP**.

Fine-grained alignment improves classification accuracy.

Table 1: Top-1 accuracy(%) of zero-shot image classification on 12 datasets. Our FILIP can boost 3~5% accuracy on average.

	CIFAR10	CIFAR100	Caltech101	StanfordCars	Flowers102	Food101	SUN397	DTD	Aircrafts	OxfordPets	EuroSAT	ImageNet	Average
CLIP-ViT-B/32	91.3	65.1	87.9	59.4	66.7	84.4	63.2	44.5	21.2	87.0	49.4	63.2	65.3
FILIP _{base} -ViT-B/32	86.9	65.5	91.9	55.4	85.3	82.8	69.1	49.3	57.2	88.1	49.9	68.8	70.9 ^{+5.6}
CLIP-ViT-L/14	96.2	77.9	92.6	77.3	78.7	92.9	67.7	55.3	36.1	93.5	59.9	75.3	75.3
FILIP _{large} -ViT-L/14	95.7	75.3	93.0	70.8	90.1	92.2	73.1	60.7	60.2	92	59.2	77.1	78.3 ^{+3.0}

Zero-Shot Classification

FILIP achieves state-of-the-art performances under all metrics on both Flickr30K and MSCOCO datasets, **except for zero-shot text-to-image retrieval on Flickr30K**.

Table 2: Results of zero-shot image-text retrieval on Flickr30K and MSCOCO datasets. The last two rows (marked with *) report the zero-shot results on Flickr30K dataset of model fine-tuned on MSCOCO dataset, following the setting of ALBEF ([Li et al., 2021a](#)).

	Flickr30K						MSCOCO					
	image-to-text			text-to-image			image-to-text			text-to-image		
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
Unicoder-VL	64.3	85.8	92.3	48.4	76.0	85.2	—	—	—	—	—	—
ImageBERT	70.7	90.2	94.0	54.3	79.6	87.5	44.0	71.2	80.4	32.3	59.0	70.2
UNITER	83.6	95.7	97.7	68.7	89.2	93.9	—	—	—	—	—	—
CLIP	88.0	98.7	99.4	68.7	90.6	95.2	58.4	81.5	88.1	37.8	62.4	72.2
ALIGN	88.6	98.7	99.7	75.7	93.8	96.8	58.6	83.0	89.7	45.6	69.8	78.6
FILIP	89.8	99.2	99.8	75.0	93.4	96.3	61.3	84.3	90.4	45.9	70.6	79.3
ALBEF*	94.1	99.5	99.7	82.8	96.3	98.1	—	—	—	—	—	—
FILIP*	95.4	99.8	100.0	84.7	97.0	98.7	—	—	—	—	—	—



06

Ablation Studies

Ablation Study

As seen all three components are beneficial for both tasks. Despite the simple design, cross-modal late interaction brings significant performance improvements over the baseline

Table 4: Ablation study of different components on pre-training subset of YFCC100M. I2T and T2I are abbreviations for image-to-text and text-to-image retrieval, respectively. “ZS” means zero-shot performance. Underlined numbers have the highest improvements for the corresponding metrics.

Model	MSCOCO				ImageNet
	I2T R@1	I2T R@5	T2I R@1	T2I R@5	ZS Top1
Baseline (ViT-B/32)	25.0	49.5	14.7	34.7	30.4
w/ image augmentation	26.1	51.8	16.5	37.5	32.5
w/ back translation	29.2	55.0	17.9	39.8	33.9
w/ cross-modal late interaction	<u>30.5</u>	<u>55.3</u>	<u>18.5</u>	<u>40.0</u>	<u>34.3</u>
Our FILIP _{base}	33.4	60.1	23.0	46.2	37.8

Ablation Study

Different settings are tried.

** denotes **the final setting** used in other experiments, but it is **not the best one**. Yet, it got **much fewer memory** requirement and **much fewer training time**.*

Loss	Embed dim	Embed precision	Token %	Training time (sec/iter)	Memory (MB)	ImageNet ZS Top1
orig (baseline)	512	fp32	-	1.31	14300	30.4
late	512	fp32	100%	2.85	26000	34.6
late	512	fp16	100%	2.67	23468	34.5
late	256	fp16	100%	2.31	22382	35.2
late	256	fp16	50%	1.61	16336	34.5
late*	256	fp16	25%	1.39	16100	34.3

Ablation Study

Component

Image augmentation

Back-translation

Late interaction

Effect

improvement

improvement

largest improvement

Results show **cross-modal late interaction gives significant gains.**

07

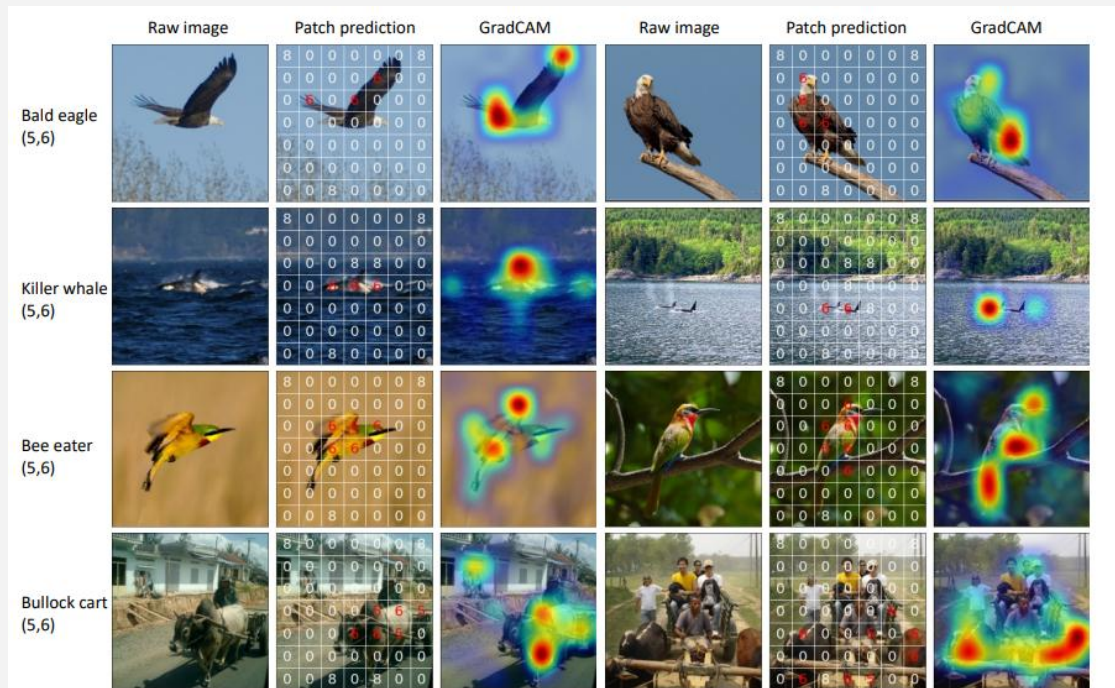
Visualization

Patch Understanding

This figure shows how the model identifies important image patches.

The patch prediction grid shows which **patch tokens** are responsible for the prediction.

GradCAM shows the activation regions from CNNs.



The model learns to focus on **semantically meaningful regions**, not random background.

FILIP vs CLIP

This figure compares FILIP with CLIP.

FILIP correctly identifies the image patches corresponding to the object label.

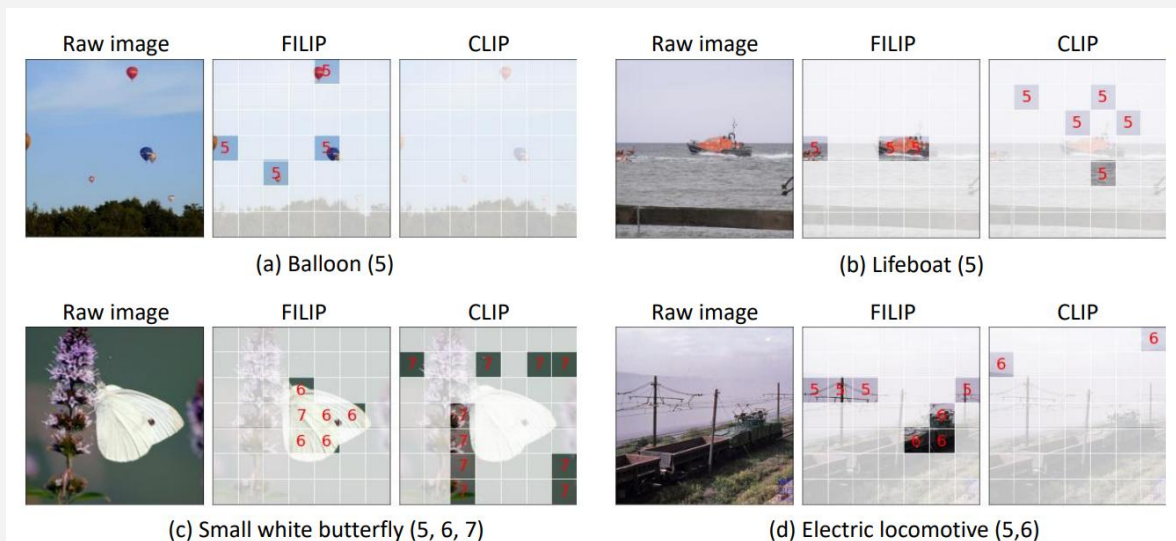


Figure 2: Visualizations of word-patch alignment for 4 classes of the ImageNet dataset and “a photo of a {label}.” is the prompt. Numbers in the parentheses after the class label indicate the location indices of the class label in the tokenized textual sequence. The correct predictions are highlighted by opaque patches with the class label indices in red.

This demonstrates that FILIP learns **fine-grained word-patch alignment** even without bounding box supervision.

Thank You!
